Programming .NET

Paolo Burgio
paolo.burgio@unimore.it

UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

High Performance
Real Time Lab


JAVA
He does exactly what I do
C#
But better.
imgflip.com

Before .NET framework (and dotNET core)

Windows GUI development

- Win32 API, MFC, Visual Basic, COM

Web (with MS technologies)

- ASP / ASP.NET

C, C++...

- Embedded

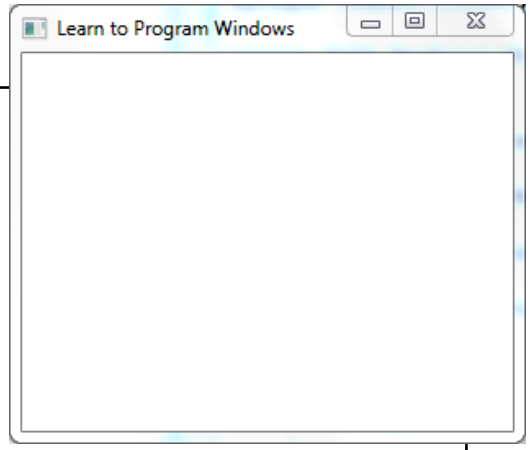
Java

- Nearly everything

Today, also

- Python for prototyping
- Flutter/react/JS derivatives for Web
- RUST for concurrent programming..

Win32 API example




```
#include <windows.h>

int WINAPI wWinMain(HINSTANCE hInstance, HINSTANCE, PWSTR pCmdLine,
                    int nCmdShow) {
  // Register the window class.
  const wchar_t CLASS_NAME[] = L"Sample Window Class";
  WNDCLASS wc = { };
  wc.lpfnWndProc = WindowProc;
  wc.hInstance = hInstance;
  wc.lpszClassName = CLASS_NAME;
  RegisterClass(&wc);
  // Create the window.
  HWND hwnd = CreateWindowEx( 0, // Optional window styles.
                              CLASS_NAME, // Window class
                              L"Learn to Program Windows", // Window text
                              WS_OVERLAPPEDWINDOW, // Window style
                              // Size and position
                              CW_USEDEFAULT, CW_USEDEFAULT, CW_USEDEFAULT, CW_USEDEFAULT,
                              NULL, // Parent window
                              NULL, // Menu
                              hInstance, // Instance handle
                              NULL // Additional application data );
  if (hwnd == NULL) { return 0; }
  ShowWindow(hwnd, nCmdShow);
  // Run the message loop.
  MSG msg = { };
  while (GetMessage(&msg, NULL, 0, 0))
   { TranslateMessage(&msg); DispatchMessage(&msg); }

  return 0;
}
```

C Win32 API

"Traditional SW development", with all its limitations we know

- Explicit memory management, pointers, malloc/free
- Functional/procedural approach, no object-oriented
- Win32 API has 1000s of global functions and datatypes

C++/MFC

C++ is an object-oriented layer built on top of C

› Can mix the two!!

Microsoft Foundation Classes (MFC) is a set of C++ classes to build Win32 applications

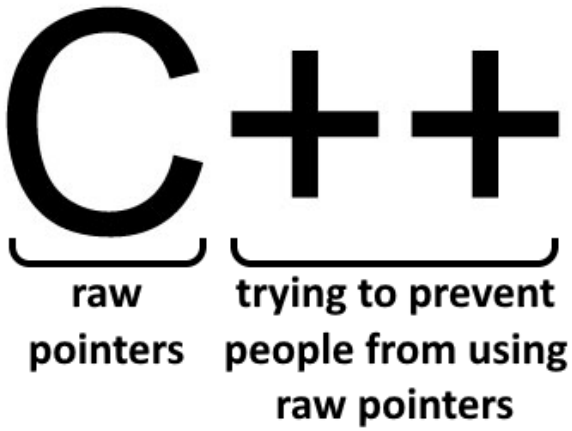
› With a lot of tools, e.g., for code-generation

› GUI?

Greatly helps, but…

› Still, enormous

› Error-prone



Java

Pros

- Solves many issues of C++
- Provides "packages" with helpful classes/functionalities
- Can embed hightly-optimized C code with JNI

Cons

- Limited capability of accessing non-Java APIs
- Little support for real cross-language

Microsoft COM

Application development framework

- Reusable binary code, e.g., between different languages
- C++ can be used in VB6
- C can be used in Delphi

Cons

- Limited language independence
- High degree of complexity
- Active Template Library (ATL) to provide C++ support classes, templates, macros..

.NET, finally

Built "after ~~failures~~ experience"

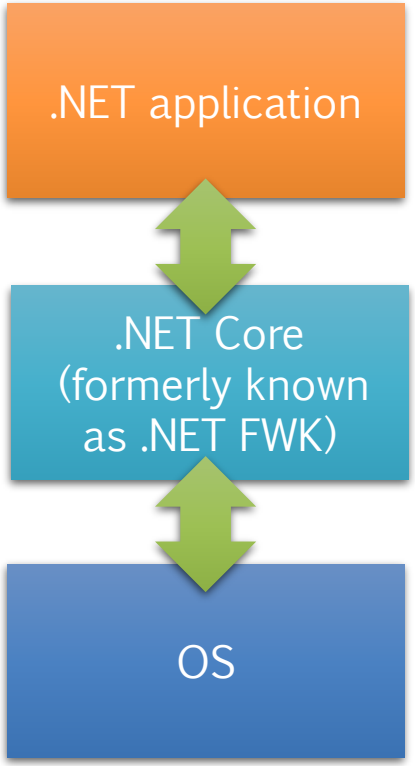
Key aspects

- OOP
- Runtime environment (MS "JVM")
- GUI
- Web
- Component-based design
- Multi-tier design
- Cross-language interoperability

**disclaimer: its logo changes quite often!*

What is .NET

- A framework
- A programming methodology
- Platform independent
- Language-insensitive



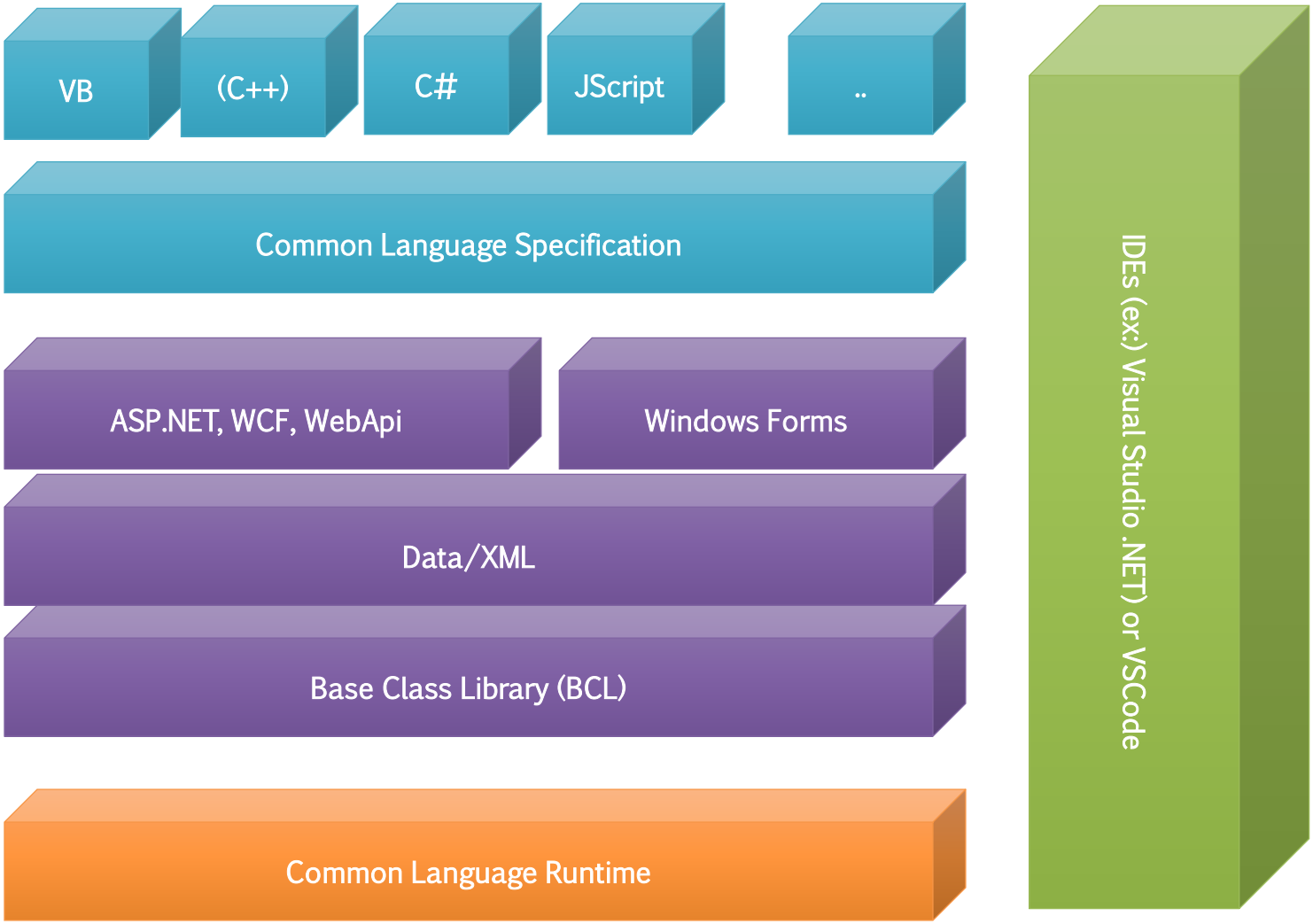
What .NET provides

- An integrated environment
- For internet | desktop | mobile app development
- Object-oriented
- Portable
  - .NET Core runs on GNU/Linux, and provides shell tools
- Managed (???)

.NET architecture

- Multi-language, cross-platform
- "JVM-like" environment called Common Language Runtime (CLR)
- Framework Class Library (FCL) providing a rich set of classes, templates…
- Just-in-Time conversion between CLR and "native" binary
- .NET components are packaged in assemblies

.NET technical architecture



.NET ecosystem

Language interoperability

- Common Language Specification (CLS)
- Common Language Runtime (CLR)
- Language-specific compilers (C#, VB, Managed C++...)

Frameworks & libraries

- WebAPI, WCF, ASP.NET, Windows Forms, MAUI, Framework Class Libs (FCL)...

Design patterns

- Heavily relies on highly scalable code.
- It's nearly impossible to program it, if you don't adopt SOLID and good

IDE

- Visual Studio, VSCode,...
- Shell

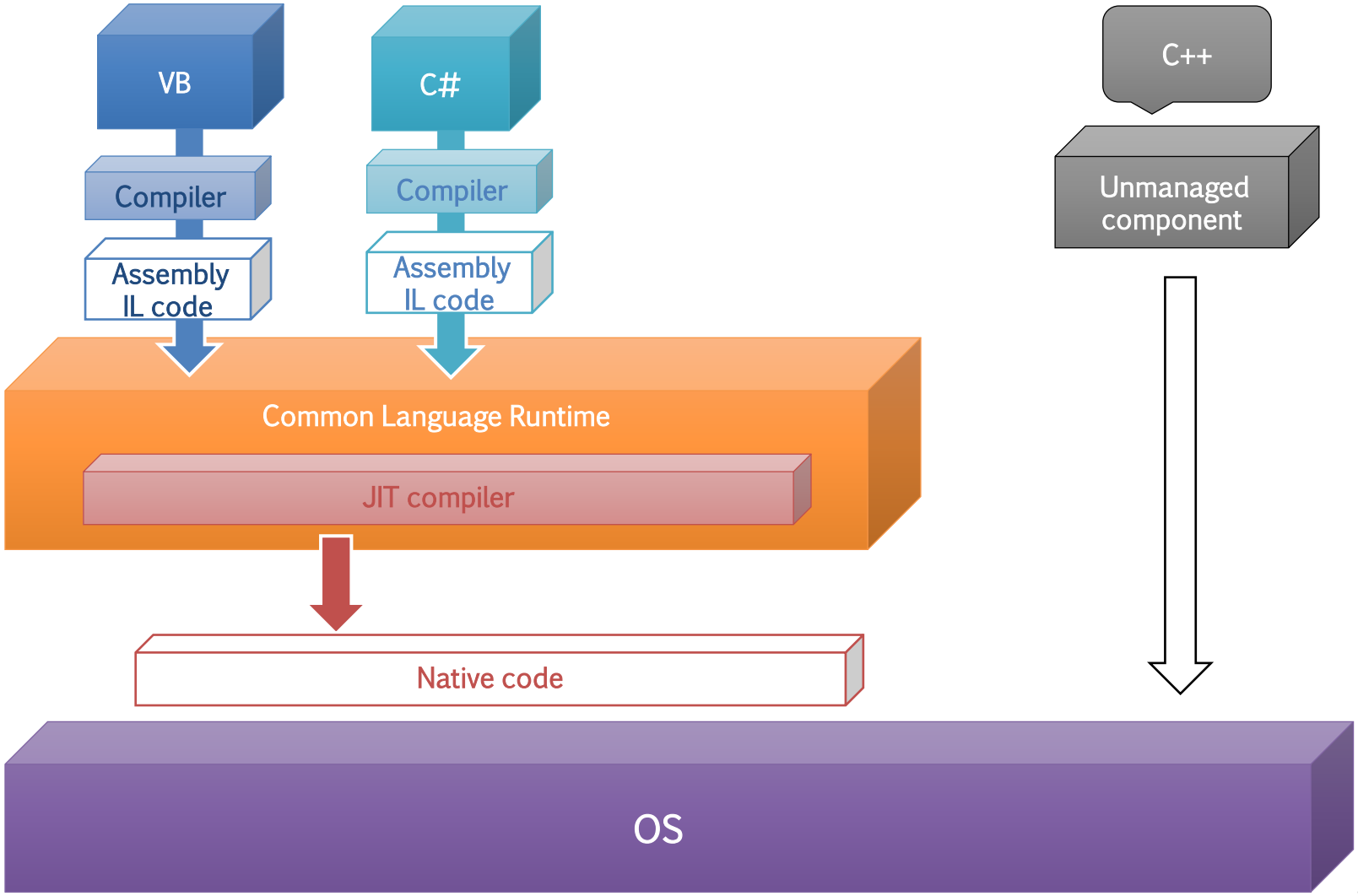
Common Language Runtime

A common runtime for all .NET languages

- Common type system
- Common metadata
- Intermediate Language (IL) to native code compilers
- Memory allocation and garbage collection

CLR Execution Model

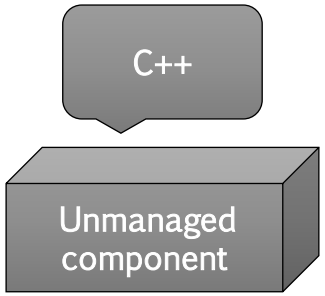
«Toolkit-based» programming



The problem with C++ compliancy

C++ has no features such as those!

- It's a «traditional» language, with syntax for classes/objects
- We call it Unmanaged, it «only» runs on OS + runtime libs
- …remember the *Toolkit* coding paradigm?



However, it is (was) still widely adopted!

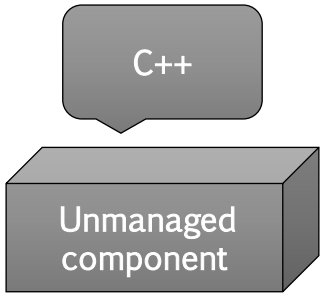
Microsoft introduced Managed C++

- Makes it possible to integrate C++ into the Framework
- They changed the C++ syntax by adding multiple keywords
- Especially with respect to…..what? ☺

The problem with C++ compliancy

C++ has no features such as those!

- It's a «traditional» language, with syntax for classes/objects
- We call it Unmanaged, it «only» runs on OS + runtime libs
- …remember the *Toolkit* coding paradigm?



However, it is (was) still widely adopted!

Microsoft introduced Managed C++

- Makes it possible to integrate C++ into the Framework
- They changed the C++ syntax by adding multiple keywords
- Especially with respect to pointers and references

Managed C++ - an example

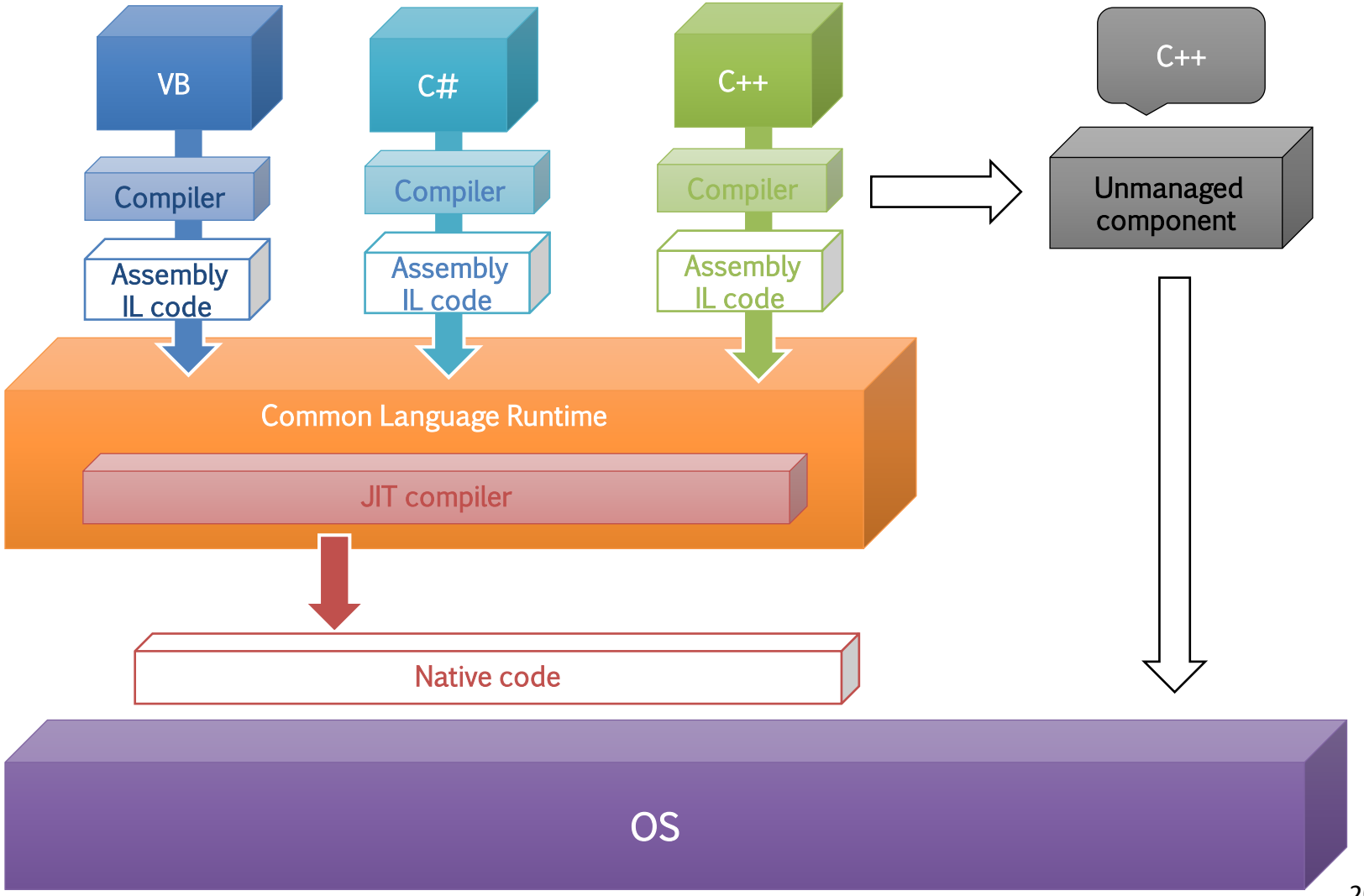
Scenario: Modena Automotive Smart Area

- Where we test&deploy autonomous driving, and smart city
- A windows server, implemented in .NET: https://github.com/pburgio/Masa.Windows
- High-perf, custom communication protocol: https://github.com/HiPeRT/MASA_protocol

I wanted to include this protocol in the server

- It's C++ based, so, I needed to make it compliant with .NET via Managed C++
- https://git.hipert.unimore.it/rcavicchioli/masa_protocol
- `feature/dotNet` branch

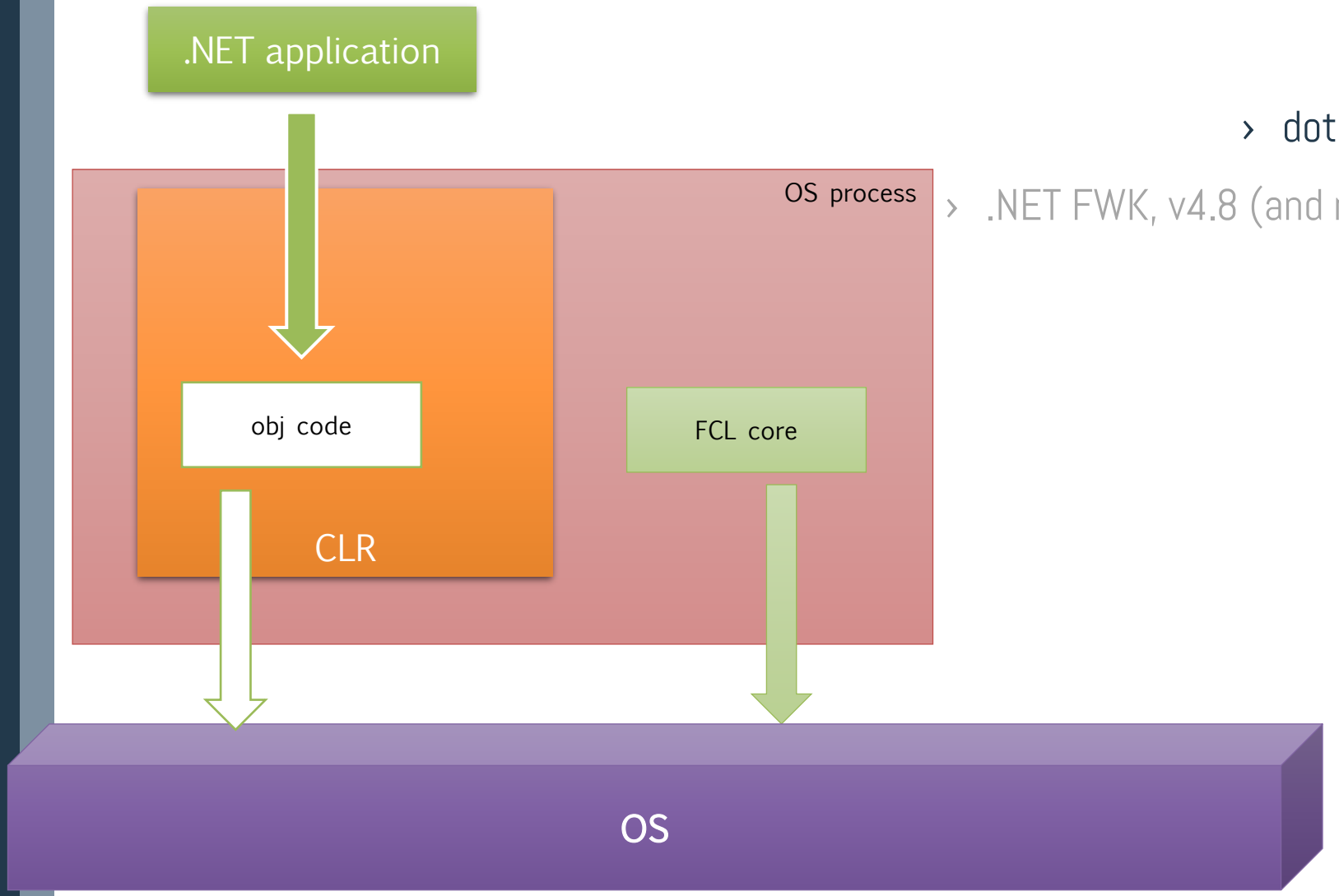
CLR Execution Model



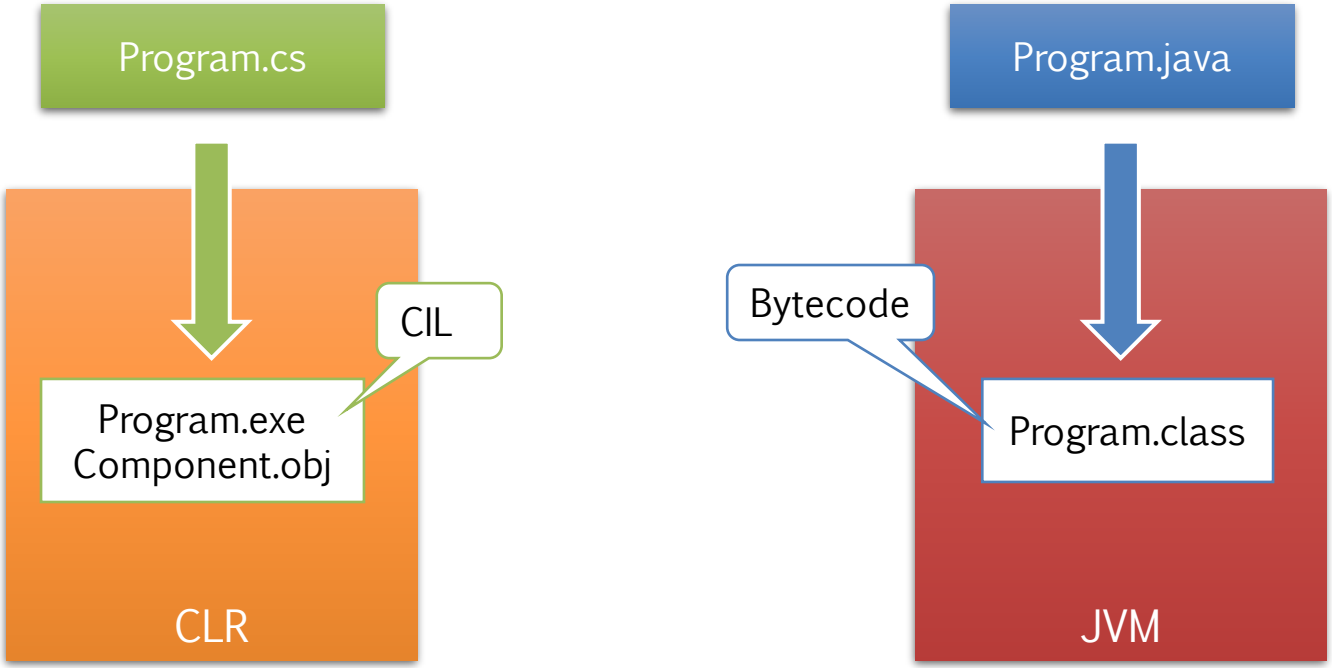
CLR Execution Model

Distributed and open source

- .NET
- dotNet Core
- .NET FWK, v4.8 (and no more)



Comparison to java



CLR and JIT compilation

"Just in time"

- All .NET languages compile to the same CIL
- The CLR transforms CIL to a given architecture ASM
- Highly-optimized process

Pros

- Support for developer services (debugging)
- Interoperation between un/managed code
- Memory management and garbage collection
- Protected "sandbox" execution
- Access to metadata (type information - reflection)

Base Class Library

- Similar to Java's System namespace
- "Core" subset of FCL used by all .NET applications
- Provides IO, Threading, DB, text, GUI, console, sockets, web…

Java vs .NET platforms

.NET is 100% proprietary (although some components are open-source), while Java is more "open"

- Mono project to provide open .NET platformn, dotNet Core was born!
- Visual Studio Code

In desktop app, Java struggled to establish its presence

- Swing, AWT (Abstract Window Toolkit), SWT (Standard Widget Toolkit)
- Also, Sun failed in promoting it properly as appealing choice for desktop products

.NET is the most popular choice for developing Windows apps

- Visual Studio Express and Community are free-of cost
- Windows doesn't ship with Java JRE; it ships with .NET

Server applications

.NET is highly integrated with Windows Server infrastructure

- E.g., integrated **Security**, Debugging, Deployment…

Server applications

- Java EE vs. ASP.NET (which is however not part of standard CLI)
- Of the top 1,000 websites, approximately 24% use ASP.NET and also 24% use Java, whereas of all the websites approximately 17% use ASP.NET and 3% use Java (http://w3techs.com/technologies/cross/programming_language/ranking)

..and mobile?

Android is based on Java

› It is also possible to develop in C (NDK)

Microsoft recently acquired Xamarin

› Cross-(mobile)platform IDE + libraries for c#

› Multi-platform APP UI «MAUI» - https://github.com/dotnet/maui

› Integrated in VS

C#

See sharp?

A bit of history…

- Born during the development of the .NET Framework, in 1999
- Team lead by Anders Hejlsberg
- Initially called "C-like Object Oriented Language" (Cool)
- When MS announced .NET project, the language had been renamed C#, and the class libraries and ASP.NET runtime had been ported to it

- Hejlsberg is C#'s principal designer and lead architect at BigM
- "Flaws in most major programming languages (e.g. C++, Java, Delphi, and Smalltalk) drove the fundamentals of the Common Language Runtime (CLR), which, in turn, drove the design of the C# language itself"

Design goals

"Simple, modern, general-purpose, object-oriented"

"Provide support for software engineering principles such as strong type checking, array bounds checking, detection of attempts to use uninitialized variables, and automatic garbage collection. Software robustness, durability, and programmer productivity are important."

And also

- Should support software for distributed environments.
- Portability (C/C++), internationalization
- No need to compete directly on performance and size with C or assembly language.

I've been talking too much...

› ...let's see it in action!

Development flow

Devtools

- Build/run based on command line
- Visual Studio vs. Visual Studio Code
- ...but basically, any editor would work

Code/folders structure

- Namespaces and assemblies (vs. Java Packages)
- Free file contents / folder structure compared to Java

Dependencies

- Nu-Get tool

Project *dotNetBasics*

Development flow (cont'd)

Solutions & projects

- References, shared projects
- Can specialize projects, e.g., for shared code/DLL, frontend, unit tests…

Config files

- Publishing, transformations
- One file for every execution environment (Prod, Dev, Local, Staging…)
- Implement the Abstract Factory pattern

Versioning

- Git (GitHub / DevOps)

.NET command-line interface (CLI)

A cross-platform toolchain for developing, building, running, and publishing .NET applications

- Comes with the .NET runtime & SDK

Create a new project (webapi or console):

```
$ dotnet new [webapi|console] [--use-controllers] [-o NAME]
```

Create a new solution:

```
$ dotnet new
```

Add the project to the solution

```
$ dotnet sln add [<PROJECT FOLDER>] # or adds current dir
```

Build:

```
$ dotnet build
```

Run:

```
$ dotnet run
```

Language: unique features

Strongly typed - more type safe than C++

- No non-safe implicit conversions
- Object-oriented (Like Java)

Local variables cannot shadow variables in enclosing non-classes blocks

- Unlike any other lang.

Arithmetic overflow automatically handled

- `checked`/`unchecked` (**statically** check blocks for arithmetic overflow)

Why are arithmetic overflow not nice?

ARIANE 5 FAILURE

- BACKGROUND:-
- European space agency's re-useable launch vehicle.
- Ariane-4 was a major success
- Ariane -5 was developed for the larger payloads

- LAUNCHED:-on June 4 1996
- MISSION was to delivered $500 million payloads to the orbit
- THE MAIDEN FLIGHT OF THE ARIANE 5 ENDED IN A FAILURE.
- ONLY AFTER 40 SECONDS THE FLIGHT VEERED OFF ITS PATH AND BROKE UP AND EXPLODED
- CAUSE: Unhandled floating point exception in code
- ENGINEERS FROM THE ARIANE PROJECT STARTED TO INVESTIGATE THE CAUSES OF LAUNCH FAILURE.

Language: basic features

- Base types and unified type system (unlike Java)
- Arrays and collections
- Pointers and references
- Implicit types (`var`)

- Boxing/unboxing
- Nullables
- Dynamic types

Object Oriented Programming

- Classes, interfaces, abstract classes, partial classes
- Properties as opposite to `get` & `set` properties
- ~~`throws`~~ `try/catch/finally` (no unchecked exceptions in Java)
- Expression bodies `<DEFINITION> => <EXPR>`

- Dynamic types
- Operator overloading (!!)
- Generic are not a type, but they are supported inside the runtime (reification)

Language: advanced features

Attributes to classes/methods

› Like Java

Functional programming

› Closures

› Delegates => lambda functions

Language-Integrated Query (LINQ)

› Query expression (SQL-like)

› Fluent expression

A few steps back...

to better understand the design principles of modern languages

Iterator
design pattern

Iterator

A **behavioral** pattern

Purpose

› Traverse elements of a collection (list, stack, tree, etc.) **sequentially** without exposing its internal representation

Motivation

› Different data structures (or even the same one) provide different algorithms for traversing them, each one with its own complexity efficiency

› Including them to the collection entity breaks its primary responsibility, which is efficient data storage

Applicability

› When you want to hide the complexity of the data structure and its (possibly) multiple access patterns

› Single class to implement traversing algorithms

› When you don't know in advance which data structure your Client should traverse

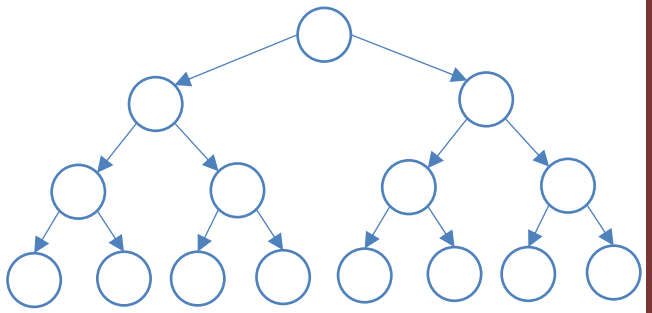
Class diagram

Project
Iterator.[Java/dotNet]

Your collection

TreeCollection

```
+ getBreadthIterator()
+ getDepthIterator()
```



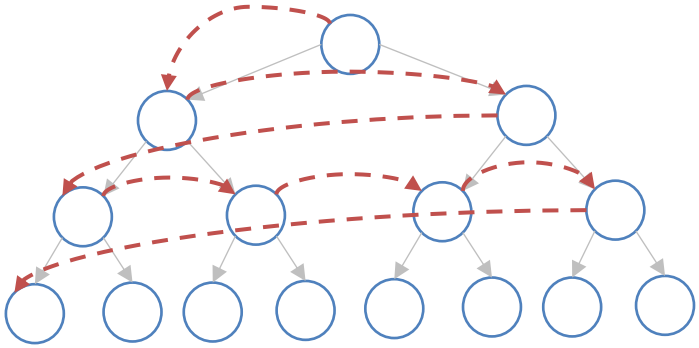
The iterator object(s)

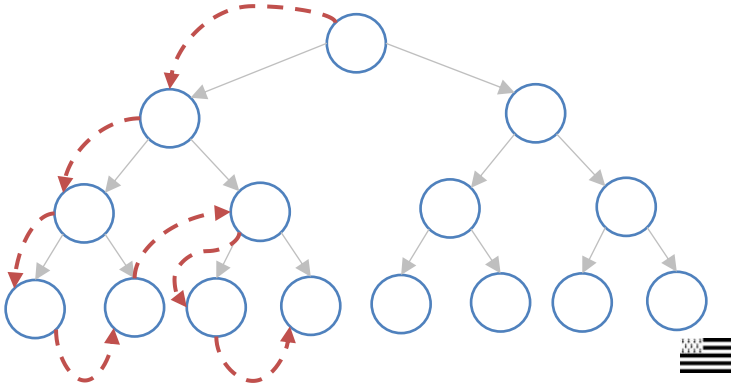
BreadthFirstIterator

```
-   currentElement
+ getNext(): Element
+ hasMore(): bool
```

DepthFirstIterator

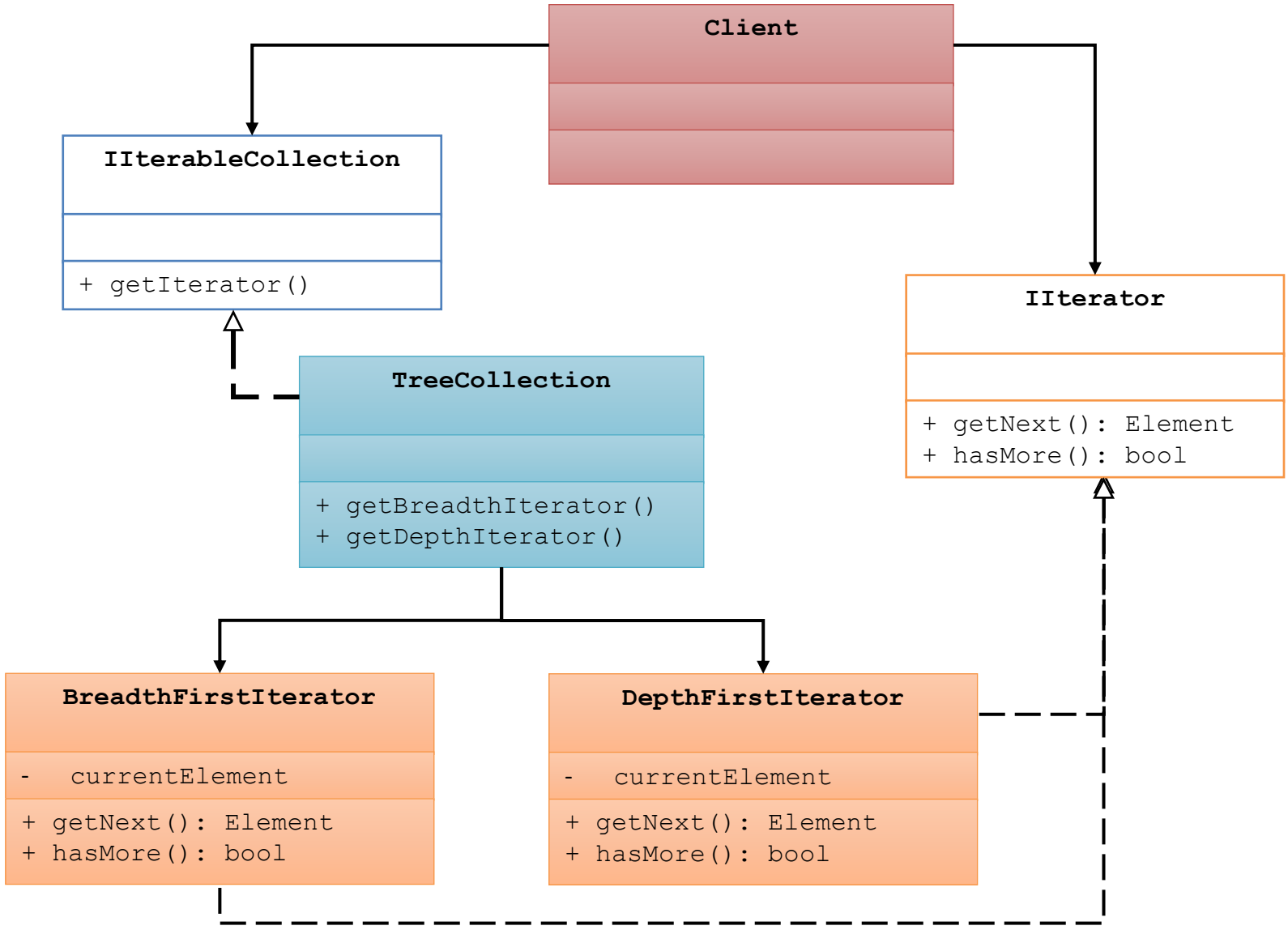
```
-   currentElement
+ getNext(): Element
+ hasMore(): bool
```





Class diagram (cont'd)

Project
Iterator.[Java/dotNet]

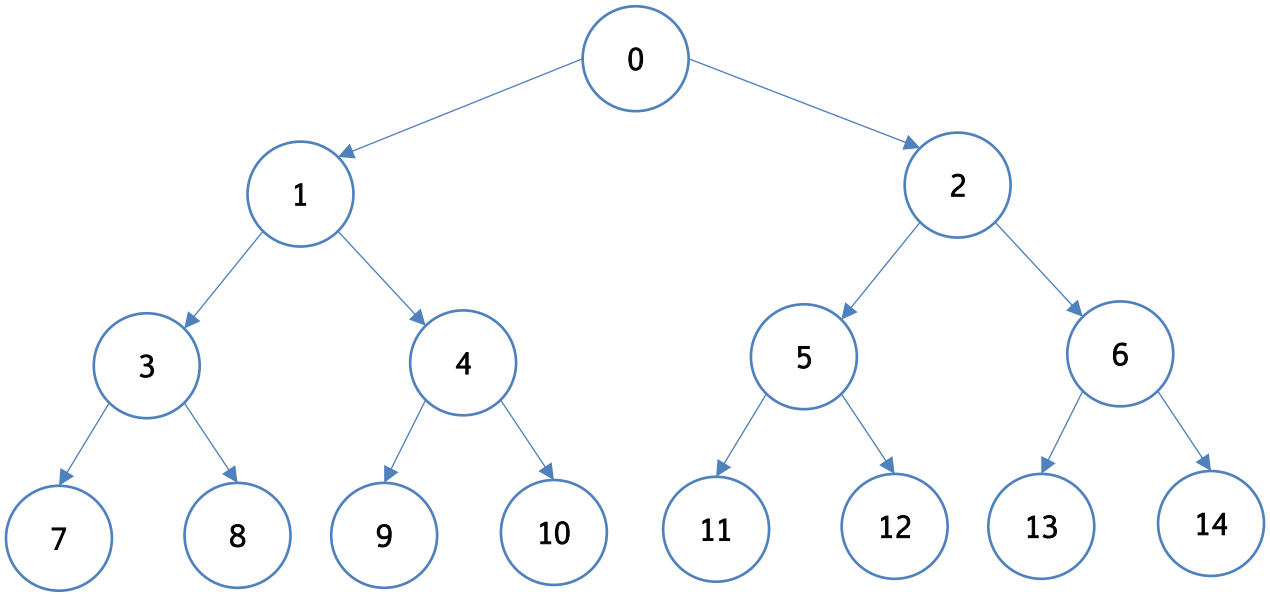


Iterator.[Java/dotNet]

Let's code!

Complete the example adding the breadth-first implementation of the traversing algorithm

- See the `DepthFirstIterator` for an example
- Solution is in the `solution` package
- The tree has the following structure



Pros/cons

Pros

- Apply SOLID
- Iterators have their internal state: you can use multiple of them concurrently

Cons

- Overkill for simple collections
- If you modify the data structure (i.e., nodes), you might reset other concurrent iterators

Visitor
design pattern

Visitor

A **behavioral** pattern

Purpose

› Separate algorithms from the objects on which they operate

Motivation

› Adheres to the Single Responsibility principle

› Makes the code more scalable

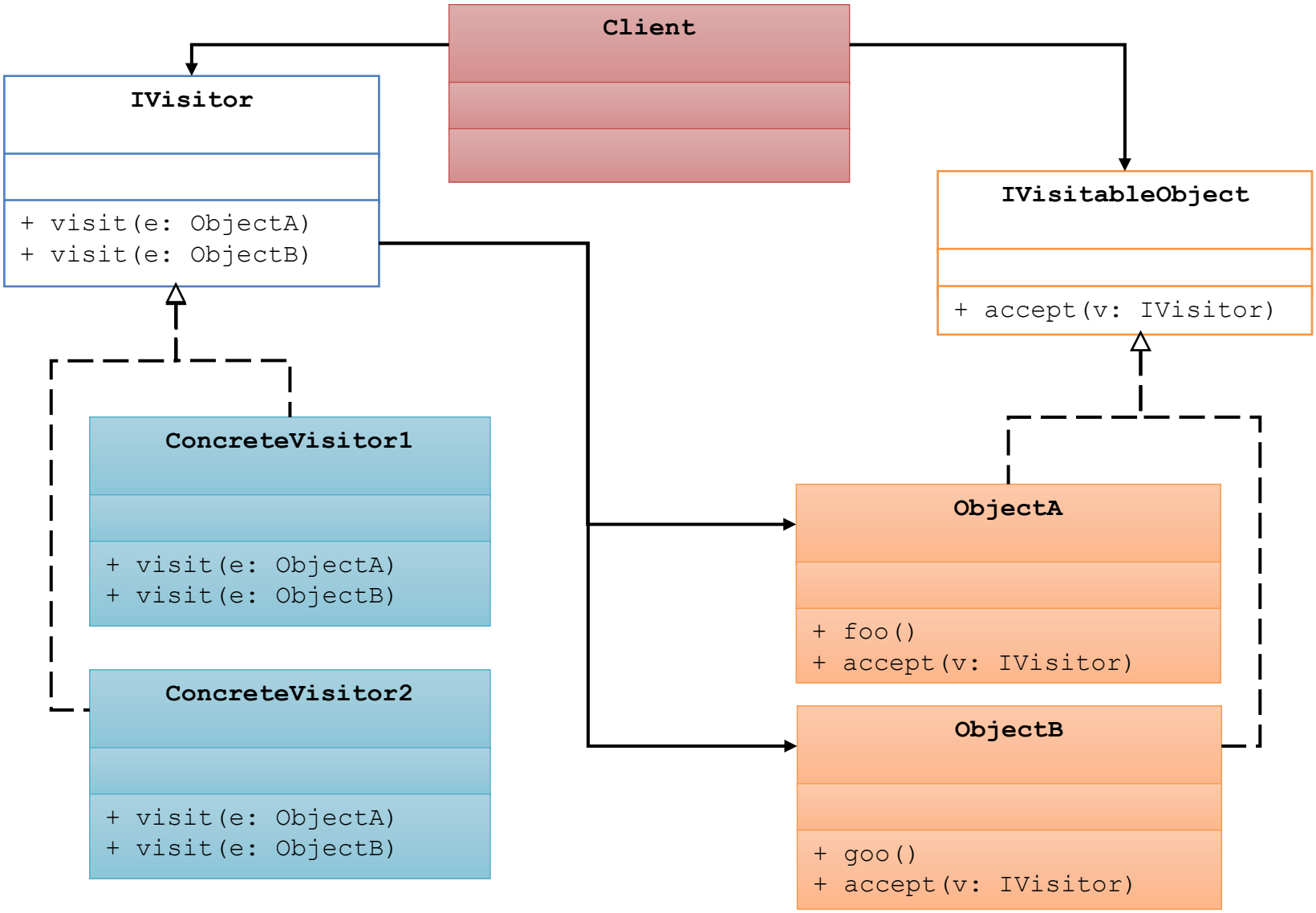
Applicability

› When you want to perform an operation on a complex data structure

› Isolate the business logic of non-business/auxiliary behaviors

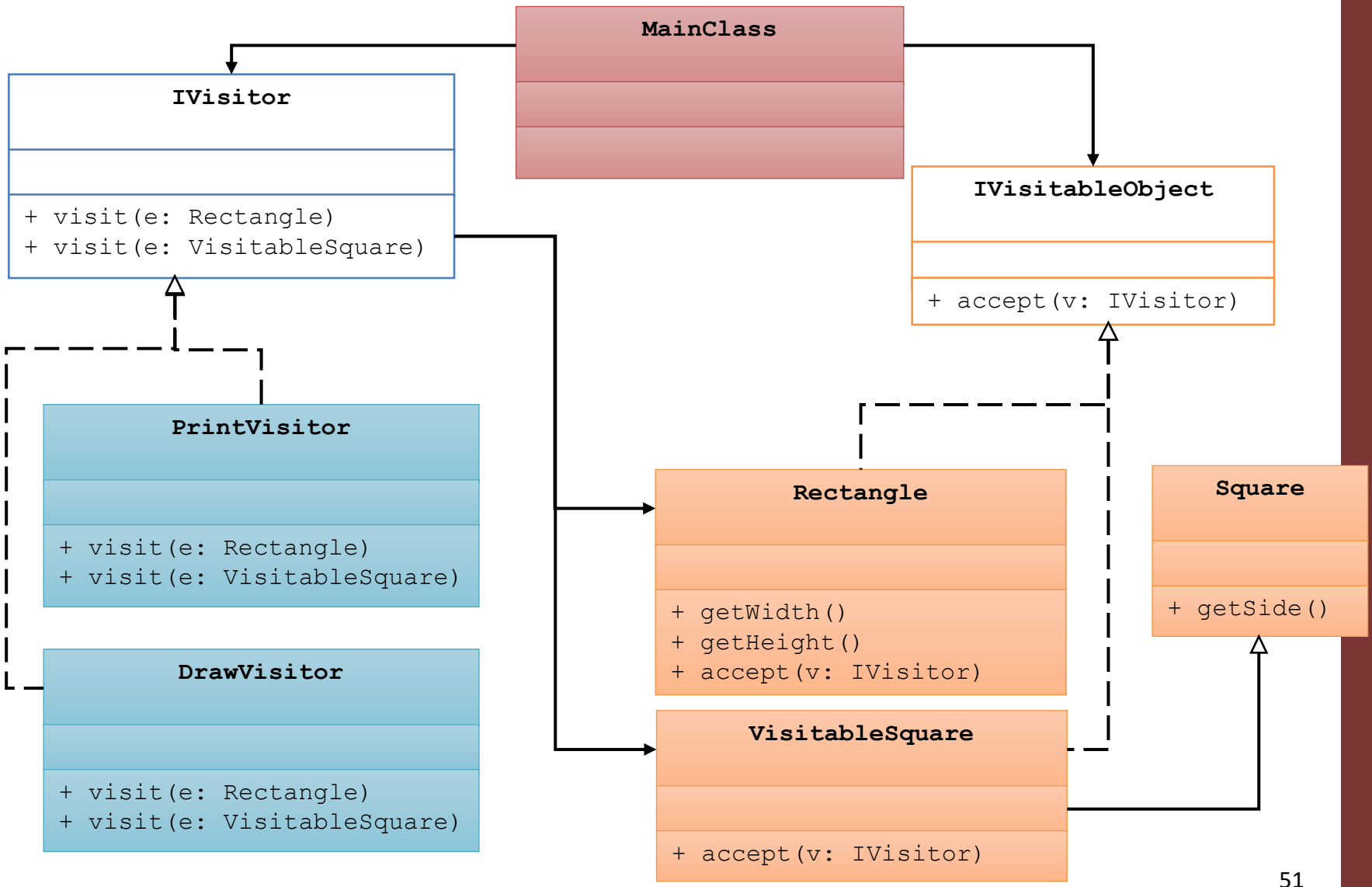
› When your behavior makes sense only in some classes of a hierarchy

Class diagram



An example

Project
Visitor.[Java/dotNet]



Pros/cons

Pros

- Apply SOLID
- Useful when applied to iterable objects (see Iterator pattern)

Cons

- Need to update visitors every time you add or remove a subclass / implementation of `IVisitableElement`
- If you modify the data structure (i.e., nodes), you might reset other concurrent iterators

Lambda expressions (recap)

Project *Lambdas.Java*

Lambda expressions are pieces of code treated as a variable

```
(a) -> System.out.println("Lambda called with " + a)
```

"A function with a parameter `a` of type integer (here, not explicit) and `printf` body"

```
//same as
public myLambda(int a) {
  System.out.println("Lambda called with " + a);
}
```

```
interface ILambda {
  void run(int a);
}
```

- You can assign/copy them…

```
ILambda myLambda = (a) -> {
  System.out.println("Lambda called with " + a);
};
```

- …of course you can invoke them….

```
myLambda.run(4);
```

- …and you can pass them as parameters ("A function within a function")

```
public class MyLambdaInvoker {
  //...
  public void invoke(ILambda action) {
    action.run(5);
  }
}
```

Fluent interface coding pattern

Not really anything new…

An Object-Oriented API / **coding pattern** that relies on method chaining

TL;DR

- "`return this`" in methods
- or return a new object which is a clone of the current one

What is a *context*, in this sense?

- The return value of method, creates a context
- In fluent interface, the context is always the same ("this")
- Ended when a method returns void

Pros/cons

Pros

- Increase code readability
- Suitable for immutables (e.g., objects whose value cannot be modified – if you return a new objects)
- Designs an interface that specifies a DSL (Domain Specific Language), identified by the context

Cons

- When mixed with lambda expressions, brings some errors at run-time, instead of compile time
- Harder to debug/log
- Might have issues when subclassing/ovrerriding fluent methods

Fluent in action

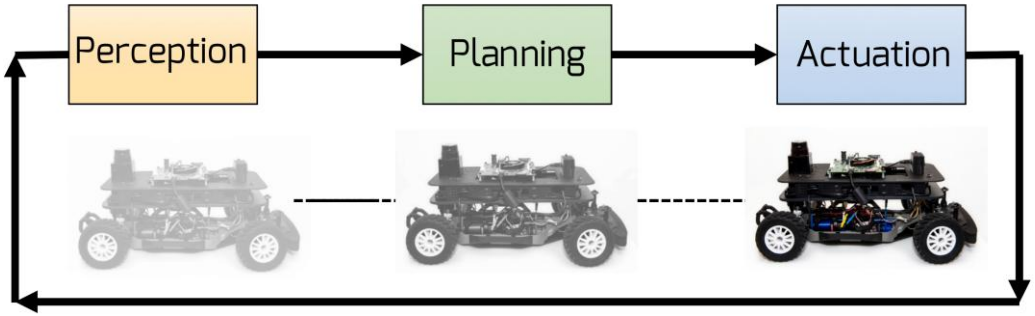
Let's code!

In the main method file

- The `StringBuilder` classes are implemented with this
  - Both for Java (`java.util.StringBuilder`) and .NET (`System.Text.StringBuilder`)
- I also implemented a hand-made example (`ADStackComposer` class)

Projects *dotNetBasics* and *Fluent.Java*

ADStackComposer: a domain-specific language




```
new ADStackComposer()
  // Set up the perception stack
  .addPerception(new YoloV8PedestrianDetector())
  .addPerception(new LidarClusteringForPedestrian())
  .addFusion(new PedestrianDetector())

  // Set up the loc
  .addPerception(new GNSSReceiver())
  .addFusion(new EKFLocalization())

  // Add maps
  .registerMap(mapsrepo.Get("Modena"))
  .registerMap(mapsrepo.Get("Reggio Emilia"))
  .registerMap(mapsrepo.Get("Mantova"))

  // Register planner, controller and vehicle backend
  .addPlanner(new PathPlanner())
  .addController(new ModelPredictiveControl())
  .addActuation(QuattroporteDBWAdapter.getInstance())

  .run();
```

Perception

Planning

Actuation

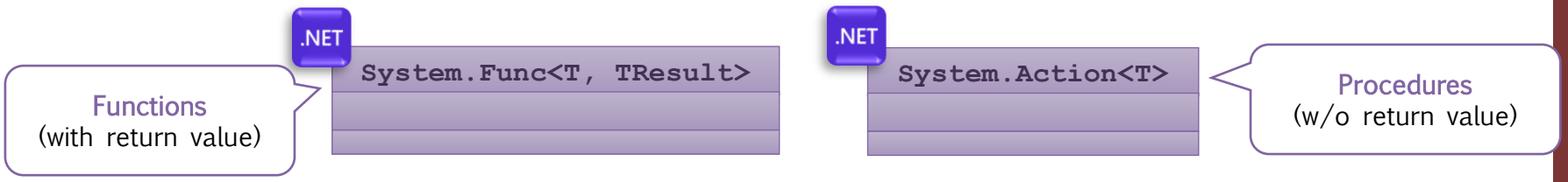
Lambda expressions in dotNet

Project *dotNetBasics*

Lambda expressions are pieces of code treated as a variable

```
(int a) => Console.WriteLine($"Lambda called with {a}");
```

"A function with a parameter `a` of type integer (here, explicit) and `WriteLine` body"



› You can assign/copy them…

```
Action<int> myLambda = (int a) =>
{
  Console.WriteLine($"Lambda called with {a}");
};
```

› …of course you can invoke them….

```
myLambda(4);
```

› …and you can pass them as parameters ("A function within a function")

```
public class MyLambdaInvoker
{
  public void Invoke(Action<int> action)
    => action(5);
}
```

Putting everything together

Language Integrated Query (LINQ)

Language-Integrated Query (LINQ)

Integration of query capabilities directly into the C# language

› Traditionally, queries against data such as SQL are expressed as simple strings
  – No type checking at compile time
  – No IntelliSense
  – No debugging

You also need to learn technologies that are specific of the data source!

› SQL databases, XML documents, …

› `for/foreach` for language constructs such as Lists, arrays, etc

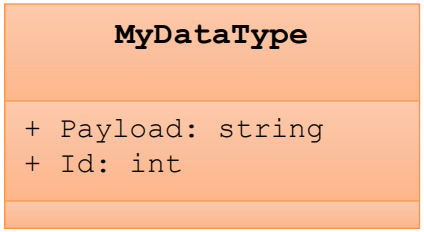
LINQ is a common solution to be applied to any data source transparently

› Decouples the access API / language

Sql-like syntax, in code

LINQ-to-objects

- For in-memory objects
- They must implement the `IEnumerable<T>` interface (the framework enforces an *Iterator* pattern)
- More complex data sources might need to implement the `IQueryable<T>` interface




```
IEnumerable<MyDataType> mylist = new List<MyDataType>();

var result = from i in mylist
             // condition
             where i.Payload.StartsWith('A')
             // AND|OR some other condition
               && i.Id > 0
             // then you can select the full object
             //  or some of its fields
             select i // …for instance i.Id

// Retrieve the first result of our query
MyDataType firstRes = result[0];
```

Everything is hidden by interfaces, even the results are `IEnumerable`

- Everything is wrapped in templates
- Here, I use generics (`var`) to let compiler infer query return type, but I could use `IEnumerable<MyDataType>`

Remote data

LINQ-to-SQL

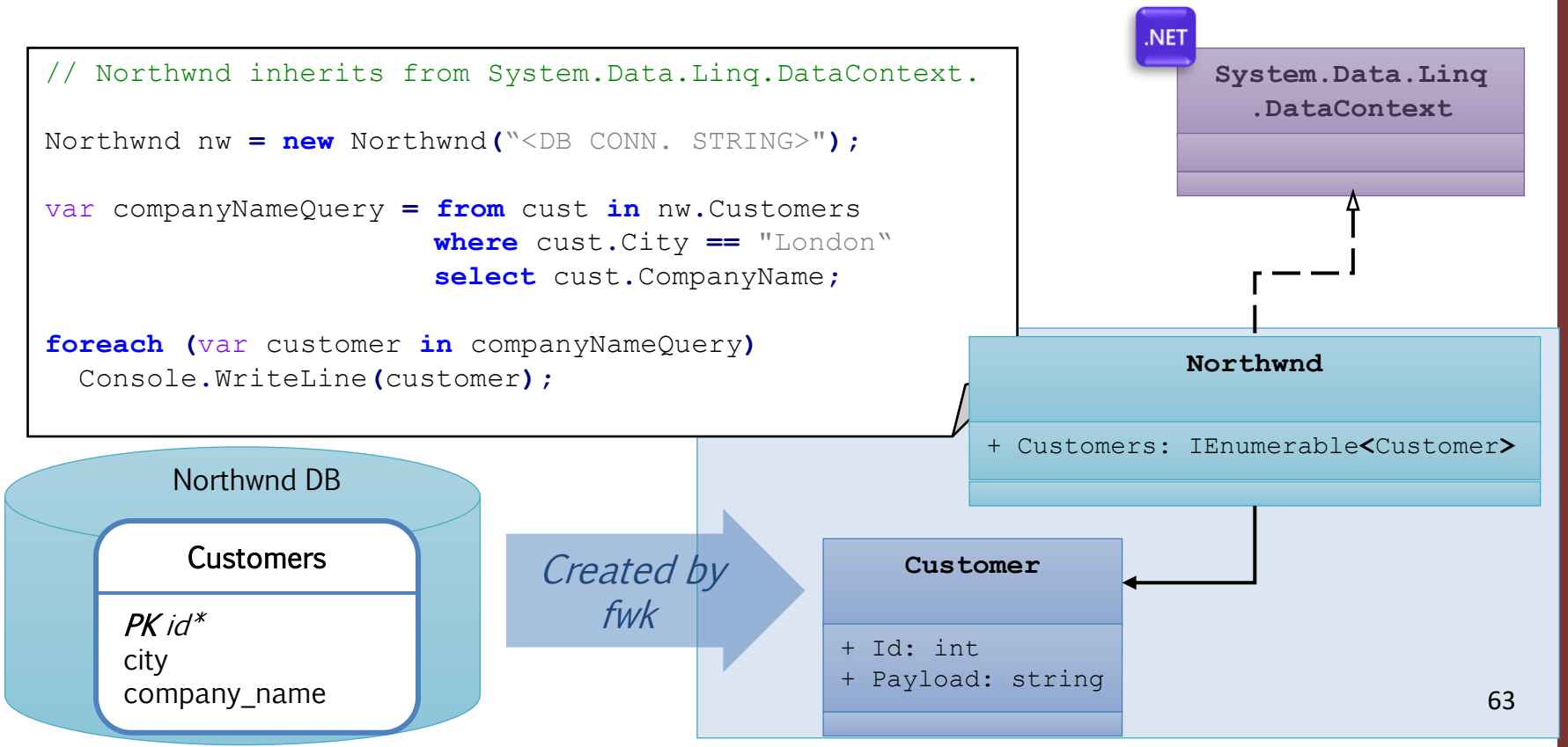
- We can use LINQ to retrieve data from DB (or even web services)!
- We query specific objects that are *created automatically* by the framework once it's bind to work with a specific DB instance
- You might want to check out the *Entity Framework (EF) Core* (this latter is **not part of the exam**)

```
// Northwnd inherits from System.Data.Linq.DataContext.

Northwnd nw = new Northwnd("<DB CONN. STRING>");

var companyNameQuery = from cust in nw.Customers
                       where cust.City == "London"
                       select cust.CompanyName;

foreach (var customer in companyNameQuery)
  Console.WriteLine(customer);
```

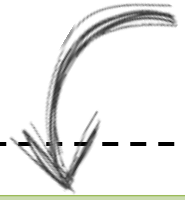


Fluent interface

- LINQ also has a fluent interface, which massively uses lambdas to enable readability and ease-of-use

```
IEnumerable<MyDataType> result = from i in mylist
                where i.Payload.StartsWith('A')
                select i.Id

MyDataType firstRes = result[0];
```



```
IEnumerable<MyDataType> result = mylist
                .Where(i => i.Payload.StartsWith("A"))
                .Select(j => j.Id)
                .OrderBy(k => k);

MyDataType firstRes = result.FirstOrDefault();
```

...or combine them thanks to the fluent approach

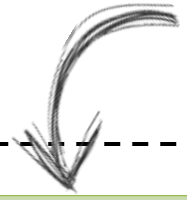
```
MyDataType firstRes = mylist
                .Where(i => i.Payload.StartsWith("A"))
                .Select(j => j.Id)
                .OrderBy(k => k)
                .FirstOrDefault();
```

Fluent interface

› LINQ also has a fluent interface, which massively uses lambdas to enable readability and ease-of-use

```
IEnumerable<MyDataType> result = from i in mylist
          where i.Payload.StartsWith('A')
     var  select i.Id

MyDataType firstRes = result[0];
```



```
IEnumerable<MyDataType> result = mylist
          .Where(i => i.Payload.StartsWith("A"))
     var  .Select(j => j.Id)
          .OrderBy(k => k);

MyDataType firstRes = result.FirstOrDefault();
```

...or combine them thanks to the fluent approach

› Hint: use generics/`var` to infer query return type of the

```
MyDataType firstRes = mylist
          .Where(i => i.Payload.StartsWith("A"))
  var     .Select(j => j.Id)
          .OrderBy(k => k)
          .FirstOrDefault();
```

Fluent LINQ language

"Designs an interface that specifies a DSL (Domain Specific Language), identified by the context" (cit. myself, a few slides ago)

In this case, the language API is inspired by SQL

- *Select*
- *Where*
- *OrderBy[Desc]*
- *First[OrDefault]*
- *Single[OrDefault]*
- *Distinct*
- *.....*

..and can even be extended!

```
MyDataType firstRes = mylist
                .Where(i => i.Payload.StartsWith("A"))
                .Select(j => j.Id)
                .OrderBy(k => k)
                .FirstOrDefault();
```

References

Course website

- http://hipert.unimore.it/people/paolob/pub/ProgSW/index.html
- https://learn.microsoft.com/en-us/dotnet/core/tools/
- Gamma, et.al «Design Patterns – Elements of reusable Object Oriented Software», Addison Wesley
- https://refactoring.guru/

How to run the examples

- Use Visual Studio IDE

...or (better)

```
$ dotnet new [webapi|console] [--use-controllers] [-o NAME]
$ dotnet build && dotnet run
```

My contacts

- paolo.burgio@unimore.it
- http://hipert.mat.unimore.it/people/paolob/